

Kaveen Sheron Jayasinghe

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Education

University of Moratuwa, Sri Lanka 2024 – Present

B.Sc. Engineering (Hons) in Computer Science & Engineering, specializing in Cyber Security.

St. Joseph's College, Colombo-10 2009 – 2023

General Certificate Exam Advanced Level, Physical Sciences stream. Z-Score 2.2905, with 3 Distinction passes.

General Certificate Exam Ordinary Level, 9 Distinction passes.

Professional Experience

Training Co-Lead, Oppia (Google-backed Open-source Foundation) 2026 – Present

Angular, Python, Express, Playwright

- Contribute to full-stack feature development, develop e2e frontend and backend tests, mentor and onboard new contributors, and coordinate quarterly project planning.

Associate Software Engineer, ictfromabc.com 2025 – Present

Laravel, SvelteKit, Flutter

- Built and integrated full-stack features; improved platform reliability by writing backend tests and expanding CI workflows for a platform serving 55,000+ users.

Projects

Prypiatos — Smart Energy Management System Prypiatos/repositories

Terraform, Ansible, Kubernetes (k3s), Argo CD, AWS (EC2, EBS), MQTT

- Provisioned a 3 node self managed k3s cluster on AWS with Terraform, defining EC2 instances and EBS storage as code.
- Automated cluster bootstrap with Ansible, covering control plane setup and worker node joining end to end.
- Implemented a GitOps deployment workflow with Argo CD, continuously syncing Git managed manifests with live cluster state.
- Built MQTT-based edge workflows for real-time energy data acquisition, with local buffering, reconnection handling, and reliable device-to-cloud delivery.

Cross-Lingual Speech & Identity Synthesis System sheronjay/CrossLingual-FaceDub

Python, WhisperX, NLLB-200, Coqui TTS, Wav2Lip, Voice Conversion

- Built an end-to-end English-to-Sinhala multimedia pipeline chaining transcription, machine translation, speaker-preserving voice conversion, and lip synchronization while trying to preserve the original speaker's voice.

Veri5 — Healthcare Tracker & Social Support Tool for STD Patients sheronjay/Syncode-Veri5

Next.js, React, TypeScript, Prisma, PostgreSQL, TensorFlow.js, JWT, OAuth

- Architected a full-stack platform with modular APIs for authentication, test-kit verification, appointments, orders, clinics, and practitioner workflows.
- Built a client-side validation pipeline with TensorFlow.js and custom EXIF parsing to run image inference in-browser, flag screenshots or edited uploads, and keep raw images off the backend.
- Designed transactional backend workflows with JWT and Google OAuth, hashed one-time status-share tokens, serial-based kit verification, and atomic updates across kit usage, verification state, and appointment availability; selected as a Finalist at Devthon 3.0.

Autonomous Drone Platform for IESL-RoboGames

Docker, Pixhawk, ArduPilot, Raspberry Pi, SSH

[IESL-RoboGames](#)

Ohana Note — Secure Message-Sharing Platform with a Game Interface

KAPLAY, Node.js, Express, MySQL, Docker Compose

[sheronjay/Ohana-Note](#)

N1K0 — Micromouse Robot

ESP32, Custom PCB Design, Real-Time Sensor Fusion

[github.com/N1K0-micromouse](#)

Kandypack — Supply Chain Management System

React, Express, MySQL

[sheronjay/supply-chain-management](#)

Open Source Contributions

- Actively contributing to the **OppiaFoundation**.
- Contributed to **WSO2** open source projects during Hacktoberfest 2025.

Leadership & Volunteering

Selection Committee Lead — SLIoT 2026

- Headed participant selection and evaluation for a National level IoT and innovation-focused program, ensuring fair and technical assessment standards.

Program Committee Lead — Hit the Grounds 2025

- Led the planning and execution of an industry-facing competition with participation from 20 leading companies.

Technical Committee Member — IESL-RoboGames 2026

- Contributed to developing a containerized sandbox environment for autonomous drone development using Docker, supporting Pixhawk, ArduPilot, and Raspberry Pi based workflows.

Member — AIESEC in Colombo South (2025–2026)

- Contributed to cross-cultural experience programs and internal operations within a global student-led organization.

Interests & Achievements

- Contributor: Asia-Pacific Robot Contest (2025); Winner: Hitachi High-Tech Award
- Sports: Volleyball (All-Island Champion 2019), Cricket (2012–2022)
- Finalist: Devthon 3.0 (2026)
- Finalist: CodeRally (2025)
- Finalist: CyberShield 4.0 (2025)

Technical Skills

Programming: C, C++, JavaScript, Python, SQL, Bash

Web/Frameworks: REST API, React, Angular, Playwright, Laravel, SvelteKit, Express

Cloud & DevOps: Git, Docker, Kubernetes, ArgoCD, Terraform, Ansible, CI/CD, AWS (S3, EC2)

Other Skills

Languages: English (Professional working proficiency), Sinhala (Native proficiency)